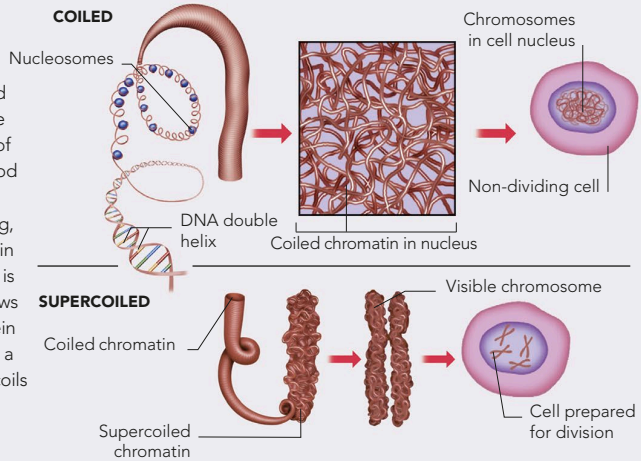


COILS AND SUPERCOILS

DNA's coiled structure allows an incredible length to be packed into a tiny space. If unwound, the DNA in a chromosome would stretch about 5cm (2in). There are 46 chromosomes in the nucleus of each cell (except mature red blood cells, which have no nucleus or DNA). When cells are not dividing, the DNA (wrapped around protein to form what is called chromatin) is relatively loosely coiled. This allows portions to be available for protein assembly and other functions. As a cell prepares to divide, its DNA coils into supercoils, which are shorter and denser, and visible as the typical chromosome "X" shapes.



DNA backbone

Constructed of alternating units of deoxyribose (a form of sugar) and phosphate chemicals

Helical repeat

DNA helix twists once for every 10.4 rungs of base pairs

Adenine-thymine link

Adenine always forms a base pair with thymine

Cytosine

Thymine

Adenine

Guanine

Guanine-cytosine link

Guanine always forms a pair with cytosine

